LAB 8- github link

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)
The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

$$\texttt{daisy}, \quad \texttt{dandelion}, \quad \texttt{roses}, \quad \texttt{sunflowers}, \quad \texttt{tulips}.$$

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.
Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.
This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

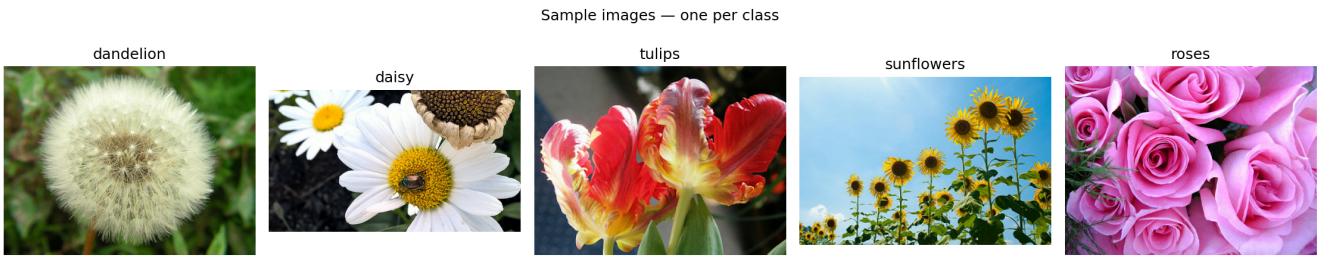


Figure 1: Sample test images, one per class (dandelion, daisy, tulips, sunflowers, roses).

Inference.

- The images differ a lot. The dandelion is a close-up on leaves, the daisy picture has a dark background and more than one flower, the tulips are large and strongly coloured, the sunflowers are small and far away against the sky, and the roses fill the whole frame. This difference in flower size and background is where attention could be expected to help.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	**Spatial Resolution**	**Channels**
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:
- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H\times W\times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:
- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1: \text{ after conv2_x},$$

$$P_2: \text{ after conv3_x},$$

$$P_3: \text{ after conv4_x},$$

$$P_4: \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	**Location**	**Resolution**	**Channels**
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:

- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H \times W \times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\max} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\max}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H \times W \times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow F_{cs}}$$

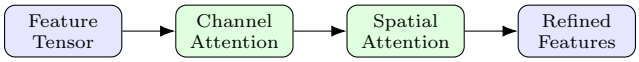
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H \times W \times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\mathrm{SA}} = F + \gamma\,\mathrm{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

$$\boxed{\text{Which spatial features are related to other spatial features?}}$$

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

Model	**Channel**	**Spatial**	**Self-Attention**
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

How does attention type affect model performance?

Students must keep the following identical:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- learning rate schedule;
- epochs;
- batch size;
- backbone;
- classifier;
- attention position; and
- evaluation procedure.

Table 1: Classification performance and computational cost for the attention-type ablation. Attention modules are inserted at P_3.

Model	Accuracy	Precision	Recall	Macro F1	Trainable params	Total params	Inference time (ms)
A0: Baseline	0.932849	0.935767	0.930507	0.932277	14,986,245	23,597,957	95.46
A1: Channel (P_3)	0.936479	0.936330	0.937676	0.935863	15,118,405	23,730,117	90.41
A2: Spatial (P_3)	0.925590	0.930434	0.922319	0.924786	14,986,344	23,598,056	109.36
A3: Channel–Spatial (P_3)	0.932849	0.934060	0.931151	0.931609	15,118,504	23,730,216	96.77
A4: Self-Attention (P_3)	0.943739	0.942798	0.944564	0.943505	16,298,246	24,909,958	115.30
A5: Combined (P_3)	0.934664	0.937527	0.931903	0.934001	16,430,505	25,042,217	97.22

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

Channel Attention → Spatial Attention

and insert exactly the same module at different stages.

Experiment	Attention Position	Resolution	Channels
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

Students should study whether attention is more beneficial for:

- high-resolution low-level features;
- intermediate representations; or
- low-resolution semantic features.

The principal question is:

How does the location of attention influence performance?

Table 2: Classification performance and computational cost for the channel–spatial attention position ablation.

Model	Accuracy	Precision	Recall	Macro F1	Trainable params	Total params	Inference time (ms)
B1: Channel–Spatial (P_1)	0.909256	0.910197	0.909487	0.908534	14,994,808	23,606,520	106.757016
B2: Channel–Spatial (P_2)	0.914701	0.918747	0.914355	0.914026	15,019,656	23,631,368	104.275908
A3: Channel–Spatial (P_3)	0.932849	0.934060	0.931151	0.931609	15,118,504	23,730,216	96.770973
B4: Channel–Spatial (P_4)	0.918330	0.920159	0.914657	0.916615	15,512,808	24,124,520	105.926776

16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.
Compare:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

As an extended experiment, students may evaluate:

$$P_2 + P_3 + P_4.$$

The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:

- parameters;
- computation;
- memory;
- training time; and
- inference time.

Students should experimentally answer:

$$\boxed{\text{Is attention at more locations necessarily more effective?}}$$

Result (multiple attention positions).
Channel–Spatial attention was tried at P_3 only, P_4 only, P_3+P_4 and P_2+P_3+P_4. The results are in the table below.

Configuration	**Accuracy**	**Macro F1**	**Total params**
P_3 only	0.9328	0.9316	23,730,216
P_4 only	0.9183	0.9166	24,124,520
P_3+P_4	0.9020	0.9013	24,256,779
P_2+P_3+P_4	0.9256	0.9249	24,290,190

Inference.

- Using attention at more places did not help. P_3+P_4 gave 0.9020, which is 3.1 points lower than P_3 alone and 1.6 points lower than P_4 alone. P_2+P_3+P_4 gave 0.9256, still 0.72 points below P_3 alone.
- The extra modules also add parameters (about 0.53 M for P_3+P_4 over P_3 alone and 0.56 M for three stages). Inference time did not change in a clear way (97.2 ms and 94.1 ms against 96.8 ms for P_3). We did not record training time or memory. Apart from P_3+P_4, the differences are small compared with the error expected on 551 test images.

17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features.
Compare:

$$\boxed{\text{Channel} \rightarrow \text{Spatial}}$$

against

$$\boxed{\text{Spatial} \rightarrow \text{Channel}}$$

while maintaining the same insertion position.
For example, at P_3:

$$F \rightarrow M_c \rightarrow M_s$$

should be compared with

$$F \to M_s \to M_c.$$

Students should determine whether selecting informative feature channels before selecting spatial regions produces behavior different from selecting spatial regions first.

Result (attention order).

Order at P_3	**Accuracy**	**Macro F1**	**Time (ms)**
Channel → Spatial	0.9328	0.9316	96.77
Spatial → Channel	0.9310	0.9299	114.68

Inference.

- The two orders have the same number of parameters and differ by 0.18 points, which is a single test image, so we cannot say that the order matters.
- Spatial→Channel had the smallest gap between training and validation accuracy (0.023) of the runs with saved histories, but this did not give a better test accuracy. It was also 18 ms slower at identical size, so the timings are noisy.

18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.

Evaluate:

$$k \in \{3, 5, 7\}.$$

Use the same attention position and training protocol.

Spatial Kernel	**Accuracy**	**Macro F1**	**Inference Time**
3×3	0.9165	0.9157	91.20 ms
5×5	0.9129	0.9119	103.93 ms
7×7	0.9256	0.9248	109.36 ms

Students should discuss whether a larger spatial kernel provides a useful increase in contextual information.

Result (spatial kernel size).

The 7×7 row is the spatial-attention model at P_3 (the default kernel). The spatial module adds 19, 51 and 99 parameters for $k = 3, 5, 7$.

Inference.

- There is no steady trend with kernel size. 7×7 was best (0.9256), but 3×3 (0.9165) did better than 5×5 (0.9129). The 3×3 model also started slowly (0.226 training accuracy in the first epoch).
- The largest difference is 1.3 points, which is within the error of one run on 551 images, so a bigger kernel did not clearly help.

19. Additional Task: Comparison of Different Attention Mechanisms

As an additional exploration task, students should compare the attention mechanisms implemented in the laboratory with established attention approaches.

Students should select at least

two additional attention mechanisms

from the following:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA);
- Convolutional Block Attention Module (CBAM);
- Coordinate Attention; and
- lightweight Self-Attention.

The pretrained backbone must remain:

$$\boxed{\text{ResNet50}}$$

for all comparisons.
To ensure a fair comparison, additional attention mechanisms should preferably be inserted at the same position:

$$\boxed{P_3 = \text{after conv4_x}}$$

unless the student provides a technical justification for using another location.

19.1 Suggested Comparison

Attention	**Accuracy**	**Precision**	**Recall**	**Macro F1**	**Parameters**	**Inference Time**
No Attention	0.9328	0.9358	0.9305	0.9323	23,597,957	95.46 ms
Channel / SE	0.9365	0.9363	0.9377	0.9359	23,730,117	90.41 ms
Spatial Attention	0.9256	0.9304	0.9223	0.9248	23,598,056	109.36 ms
ECA	0.9111	0.9105	0.9117	0.9106	23,597,962	103.11 ms
CBAM	0.9074	0.9064	0.9060	0.9059	23,730,216	106.64 ms
Self-Attention	0.9437	0.9428	0.9446	0.9435	24,909,958	115.30 ms
Additional Mechanism (Channel+Spatial)	0.9328	0.9341	0.9312	0.9316	23,730,216	96.77 ms

19.2 Questions for Attention-Mechanism Comparison

Students should investigate:
1. How does classification accuracy vary across attention mechanisms?
2. How does macro F1-score vary across attention mechanisms?
3. How many parameters are introduced by each attention module?
4. How does each attention mechanism influence inference time?
5. Does attention improve the convergence behavior?
6. Does attention produce more focused feature localization?
7. Does an increase in attention complexity always improve classification performance?
8. Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?

The objective is to understand:

$$\boxed{\text{Performance} + \text{Complexity} + \text{Attention Behaviour}}$$

rather than considering only classification accuracy.

Answers to the questions in 19.2.

1. Accuracy. Self-attention 0.9437 > channel/SE-style 0.9365 > baseline 0.9328 > spatial 0.9256 > ECA 0.9111 > CBAM 0.9074.

2. Macro F1. The order is the same: 0.9435, 0.9359, 0.9323, 0.9248, 0.9106, 0.9059. F1 stays within 0.002 of accuracy.

3. Parameters added. Channel (FC) 132,160; spatial 99; ECA 5; CBAM 132,259; self-attention 1,312,001 (baseline 23,597,957).

4. Inference time. Baseline 95.46 ms; channel 90.41; spatial 109.36; ECA 103.11; CBAM 106.64; self-attention 115.30. Only self-attention is clearly slower in line with its size. The other differences are within timing noise (the two attention orders have identical size but differ by 18 ms).

5. Convergence. Self-attention started faster (first-epoch training accuracy 0.771 against 0.745), but no attention model reached a lower final validation loss than the baseline (0.221). Channel+Spatial was the least stable when fine-tuning started (validation loss 0.42).

6. Feature localisation. The spatial attention maps at P_2, P_3 and P_4 are almost uniform over the flower and only change at the image borders (Sections 28.2 and 29). The Channel+Spatial Grad-CAM maps were taken at the spatial-attention output layer and not at the last convolution block, so they cannot be compared with the baseline maps (Section 30). We therefore cannot say that attention improved localisation.

7. Does more complexity always help? No. The largest model (Combined, 25.04 M parameters) scored 0.9347, below self-attention (0.9437), and CBAM, which has the same size as Channel+Spatial, scored 0.9074 against 0.9328.

8. Lightweight against heavy. The 132 k-parameter channel module (0.9365) did as well as or better than the 1.44 M-parameter Combined model (0.9347). ECA, with only 5 parameters, did not match it (0.9111).

20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.

The exact values selected by the student should remain identical for all controlled comparisons.

21. Loss Function

For the five-class classification problem, use categorical cross-entropy.

For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.

The same loss formulation must be used for every attention comparison.

22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

23. Required Main Result Table

Students should summarize the major experiments using the following table.

Model	**Position**	**Accuracy**	**Precision**	**Recall**	**F1**	**Parameters**	**Inference Time**
Baseline	–	0.9328	0.9358	0.9305	0.9323	23,597,957	95.46 ms
Channel Attention	P_3	0.9365	0.9363	0.9377	0.9359	23,730,117	90.41 ms
Spatial Attention	P_3	0.9256	0.9304	0.9223	0.9248	23,598,056	109.36 ms
Channel+Spatial	P_2	0.9147	0.9187	0.9144	0.9140	23,631,368	104.28 ms
Channel+Spatial	P_3	0.9328	0.9341	0.9312	0.9316	23,730,216	96.77 ms
Channel+Spatial	P_4	0.9183	0.9202	0.9147	0.9166	24,124,520	105.93 ms
Self-Attention	P_3	0.9437	0.9428	0.9446	0.9435	24,909,958	115.30 ms
Additional Attention-1 (ECA)	P_3	0.9111	0.9105	0.9117	0.9106	23,597,962	103.11 ms
Additional Attention-2 (CBAM)	P_3	0.9074	0.9064	0.9060	0.9059	23,730,216	106.64 ms

24. Graphical Performance Analysis

Students should not present only numerical tables.
Appropriate graphs must be generated to analyze model behavior.
At least three of the following graphical comparisons should be included in the laboratory report.

24.1 Training and Validation Accuracy

Plot

$$\text{Epoch} \quad \text{versus} \quad \text{Accuracy}$$

for:

- baseline ResNet50; and
- selected attention models.

Both training and validation accuracy should be shown.
Students should comment on whether attention:

- accelerates convergence;
- produces smoother learning;
- reduces the train–validation gap;
- leads to overfitting; or
- causes unstable training.

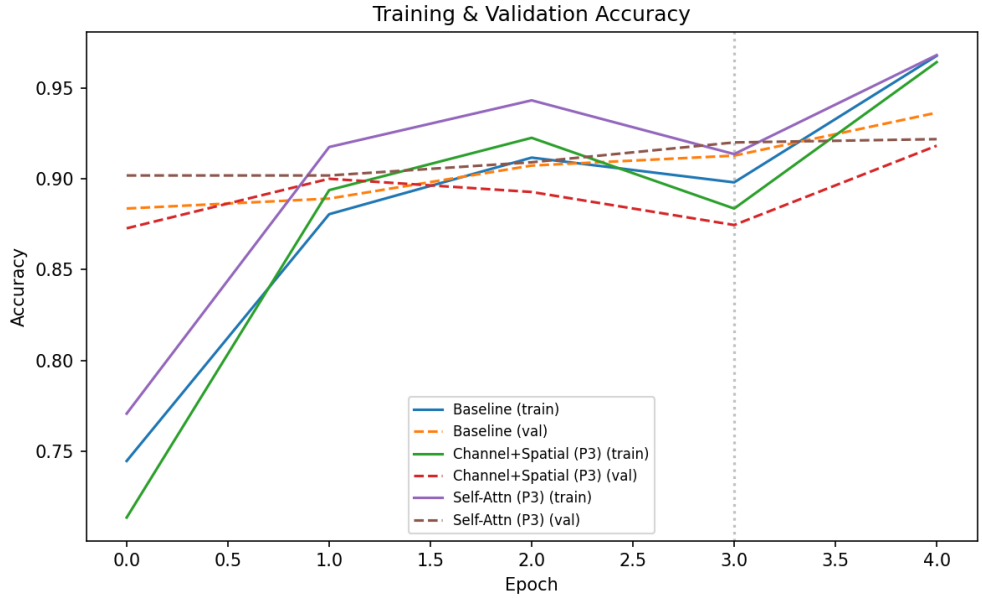


Figure 2: Training and validation accuracy of the baseline, Channel+Spatial (P_3) and Self-Attention (P_3). The dotted line marks the start of fine-tuning.

Inference.

- **Convergence:** Self-attention started higher than the baseline (first-epoch training accuracy 0.771 against 0.745, validation 0.902 against 0.884), so it learned a little faster at the start. Channel+Spatial started lower (0.714). By the last epoch all three reach about 0.965–0.968 training accuracy.
- **Smoothness:** All three curves dip when fine-tuning starts at epoch 4. The self-attention validation curve is the flattest (0.902 to 0.922), but none of the attention models is clearly smoother than the baseline.
- **Train–validation gap:** The final gap is 0.031 for the baseline, 0.046 for self-attention and 0.046 for Channel+Spatial. Attention did not reduce the gap, and the larger gap suggests slightly more overfitting. The baseline had the highest final validation accuracy (0.9364).

24.2 Training and Validation Loss

Plot

$$\text{Epoch} \quad \text{versus} \quad \mathcal{L}_{CE}.$$

The graph should include training and validation loss.

Students should compare the convergence behavior of the baseline and attention models.

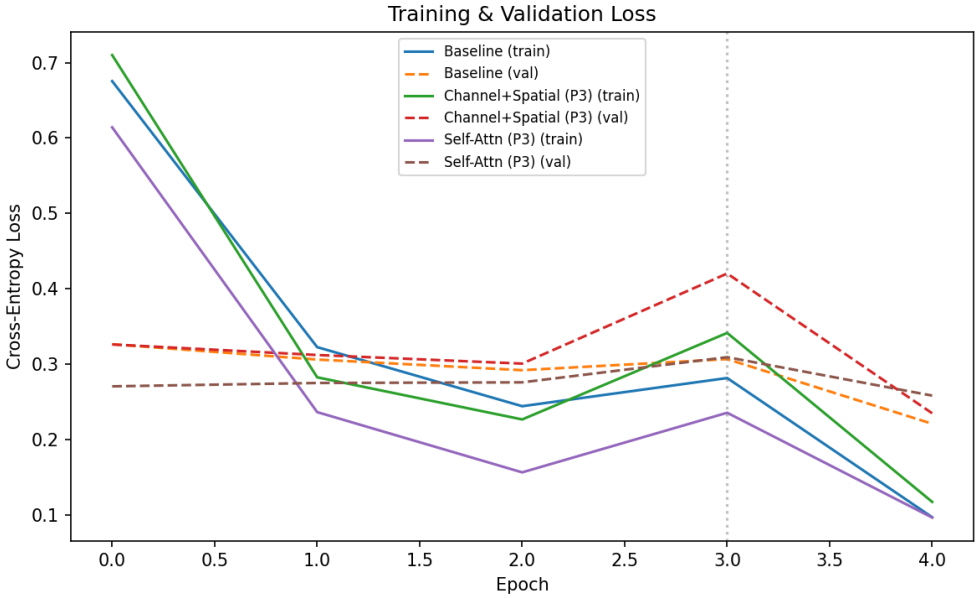


Figure 3: Training and validation cross-entropy loss for the same three models.

Inference.

- Training loss falls from 0.6–0.7 to about 0.1 for all models. The baseline has the lowest final validation loss (0.221), followed by Channel+Spatial (0.235) and self-attention (0.258), so attention did not improve the final convergence.
- Channel+Spatial is the least stable: its validation loss goes up to 0.42 when fine-tuning starts, compared with 0.307 for the baseline. The self-attention validation loss stays flat (0.27–0.31) but does not end lower than the baseline.

24.3 Attention Mechanism versus Accuracy

Generate a bar graph:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Classification Accuracy}.$$

Possible entries are:

Baseline, Channel, Spatial, Channel+Spatial, Self-Attention, SE, ECA, CBAM.

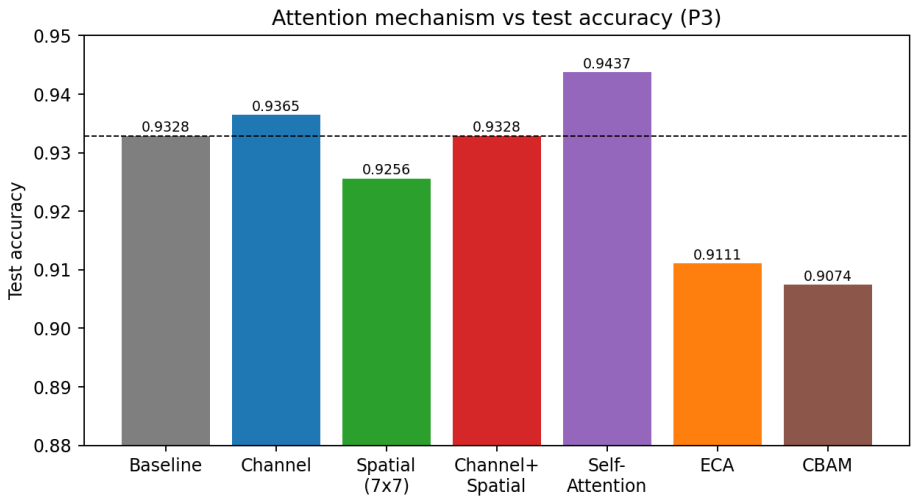


Figure 4: Attention mechanism versus test accuracy (dashed line = baseline).

Inference.

- Self-attention has the highest accuracy (0.9437), then channel attention (0.9365), the baseline and Channel+Spatial (both 0.9328), spatial (0.9256), ECA (0.9111) and CBAM (0.9074).
- Only self-attention (+1.09 points) and channel attention (+0.36 points) are above the baseline. With 551 test images the error is about ±2 points, so most of these differences are not meaningful. A McNemar test agrees: self-attention is right on 18 images where the baseline is wrong and wrong on 12 where the baseline is right ($p = 0.36$), and Channel+Spatial and the baseline differ on 14 images each way ($p = 1.0$).

24.4 Attention Mechanism versus Macro F1

Similarly, plot:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Macro F1-score.}$$

This comparison is useful when the class-wise performance is not uniform.

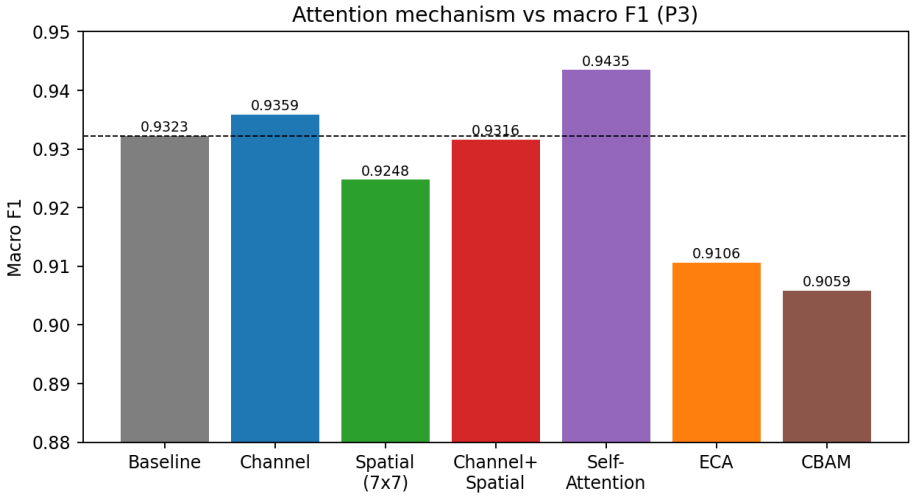


Figure 5: Attention mechanism versus macro F1 (dashed line = baseline).

Inference.

- The order is the same as for accuracy, and macro F1 is within 0.002 of accuracy for every model, so the classes are treated about equally.
- Self-attention again has the best macro F1 (0.9435, baseline 0.9323), while ECA (0.9106) and CBAM (0.9059) are the lowest.

24.5 Accuracy versus Model Parameters

Create a scatter plot using:

$$x = \text{Total Model Parameters}$$

and

$$y = \text{Test Accuracy.}$$

Each point should represent one attention mechanism.
This allows the student to study the

$$\boxed{\text{Accuracy–Complexity Trade-off}}$$

instead of evaluating accuracy independently.

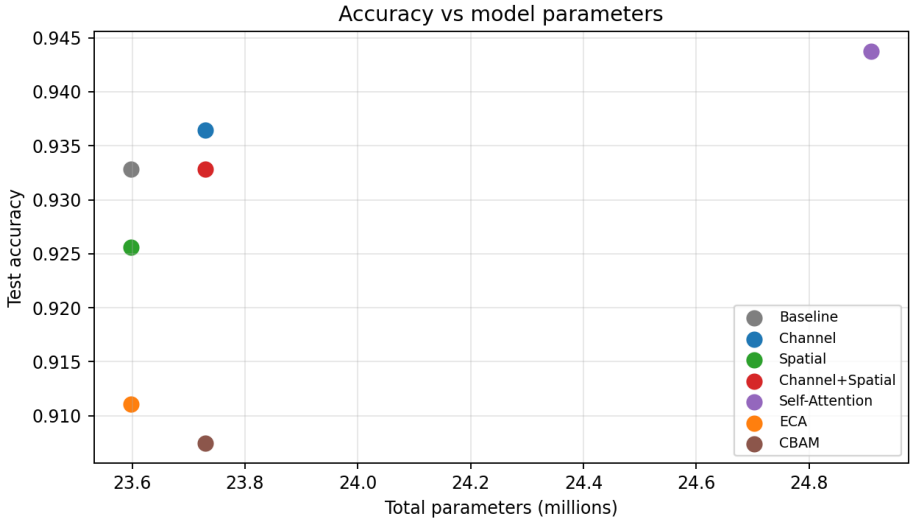


Figure 6: Test accuracy versus total model parameters (one point per mechanism).

Inference.

- Accuracy does not go up steadily with the number of parameters. Channel attention adds 0.13 M parameters for +0.36 points and self-attention adds 1.31 M for +1.09 points, while ECA (5 extra parameters) and CBAM (0.13 M) fall below the baseline.
- So more parameters do not guarantee higher accuracy. The small channel module gives a better trade-off than the heavier variants, and only self-attention gives a clearly larger gain, at a larger cost.

24.6 Macro F1 versus Inference Time

Plot:

$$x = \text{Inference Time per Image}$$

and

$$y = \text{Macro F1-score.}$$

Students should analyze whether additional computational cost produces a meaningful increase in classification performance.

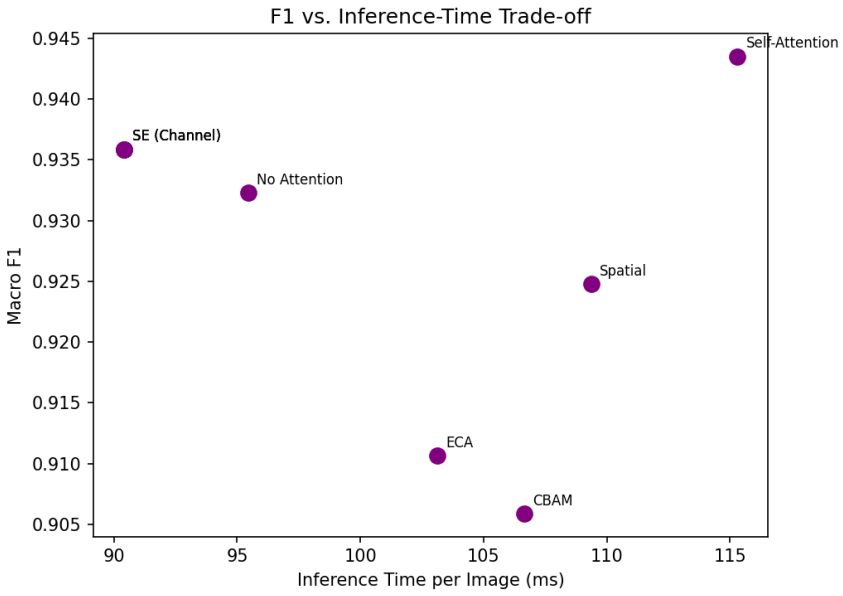


Figure 7: Macro F1 versus inference time per image.

Inference.

- Self-attention has the best F1 (0.9435) but is the slowest (115.3 ms, about 21% more than the baseline's 95.5 ms). Channel attention reaches 0.9359 at 90.4 ms, so the extra time is not needed to get a gain.
- Spatial attention, ECA and CBAM take longer than the baseline (103–109 ms) and have lower F1, so the extra time bought nothing. The timings are noisy: the two attention orders have the same size but took 96.8 and 114.7 ms.

25. Confusion Matrix Analysis

Generate a confusion matrix for:

1. baseline ResNet50; and
2. the selected attention-based ResNet50.

The confusion matrix should contain all five flower classes.

Students should analyze:

- which classes are most accurately classified;
- which classes are frequently confused;
- which class benefits most from attention;
- whether attention reduces a particular type of confusion; and
- whether the improved overall accuracy is distributed across classes.

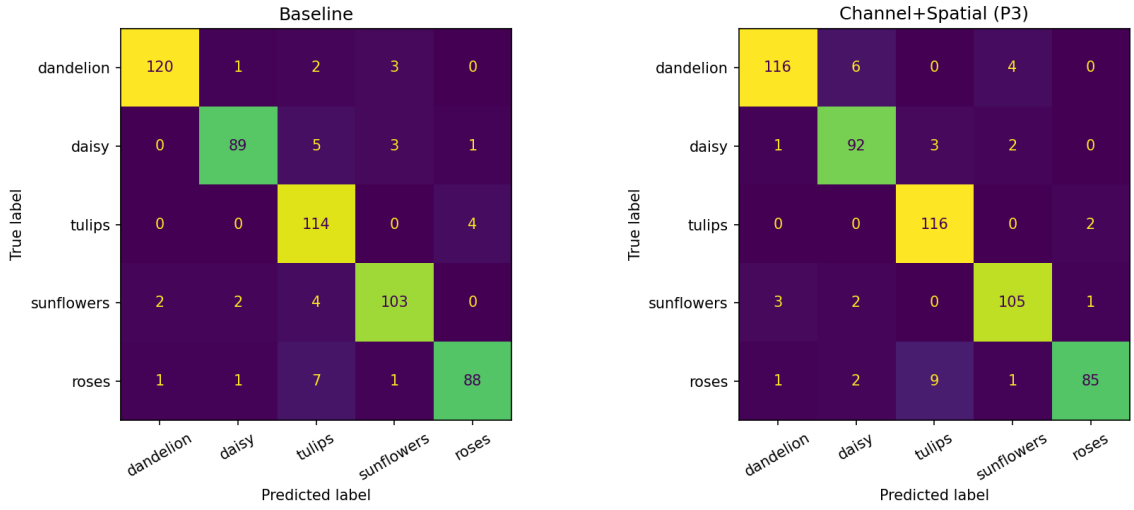


Figure 8: Confusion matrices on the test set: baseline and Channel+Spatial (P_3); both classify 514 of 551 images correctly.

Inference.

- **Best and most confused classes:** tulips are classified best (recall 0.966 for the baseline, 0.983 with attention), followed by dandelion (0.952 for the baseline). The most common confusion is roses predicted as tulips (7 images for the baseline, 9 with attention).
- **Benefit:** attention helped tulips most (F1 0.912 to 0.943), then sunflowers (0.932 to 0.942), and daisy recall went from 0.908 to 0.939.
- **Not uniform:** the overall accuracy is the same (0.9328), but the errors move around. Dandelion got worse (dandelion predicted as daisy went from 1 to 6 images; F1 0.964 to 0.939) and roses recall dropped from 0.898 to 0.867. Attention removed some tulip and sunflower errors but created others, so the total stayed the same.

26. Qualitative Analysis of Attention

Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

Students must therefore perform qualitative analysis.

The visual analysis should include:

- internal attention maps;
- attention overlays;

- intermediate feature maps; and
- Grad-CAM explanations.

Students should preferably use the same input image when comparing different models.

27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.
Let

$$A^k$$

represent feature map k from a selected convolutional layer.
For class score y^c, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The map should be resized to the original image resolution and overlaid on the input.
Students should compare:

Input | Baseline Grad-CAM | Attention Model Grad-CAM

for the same image.

27.1 Grad-CAM Interpretation

Students should qualitatively determine whether the attention model:
- focuses more strongly on the flower;
- suppresses irrelevant background regions;
- concentrates on petals;
- concentrates on the central flower structure;
- attends to discriminative edges;
- distributes activation across the entire flower;
- incorrectly focuses on leaves;
- incorrectly focuses on background objects; or
- changes its focus for an incorrect prediction.

Students should avoid unsupported statements such as:

> "The Grad-CAM of Model A is better."

Instead, the observation should describe visible evidence.
For example:

> The baseline network activates both the flower and surrounding leaves, whereas the attention-based model produces stronger activation over the petal region and weaker activation over the background.



Figure 9: Input, baseline Grad-CAM and Channel+Spatial (P_3) Grad-CAM for a dandelion test image (both predict dandelion).

Inference.

- The baseline Grad-CAM is strongest at the centre of the dandelion head and gets weaker over the white fluff, with little activation on the leaves.
- The Channel+Spatial map is weak and almost the same everywhere, with a small spot in the lower-right corner, which is background. This map was taken at the spatial-attention output layer, while the baseline map was taken at the last ResNet block (conv5_block3_out). The two cannot be compared directly, so we cannot say that the attention model localises worse.
- This is only one correct image. A fair comparison needs the attention model's Grad-CAM at its last convolution block, on more images.

28. Internal Attention Maps

An important objective of this experiment is to distinguish between:

$$\boxed{\text{Internal Attention}}$$

and

$$\boxed{\text{Post-hoc Explanation}}$$

because they are not the same.
An attention module directly computes weights during the forward pass.
Grad-CAM, in contrast, is generated after prediction using gradients.

28.1 Channel-Attention Visualization

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Students should visualize the learned channel importance values using a bar graph.
Instead of displaying every channel when C is large, the student may show the top-K channels.
Recommended values are:

$$K = 10$$

or

$$K = 20.$$

Students should comment on whether:

- a small number of channels receive strong weights;
- many channels contribute similarly; or

- channel weighting changes substantially for different input images.

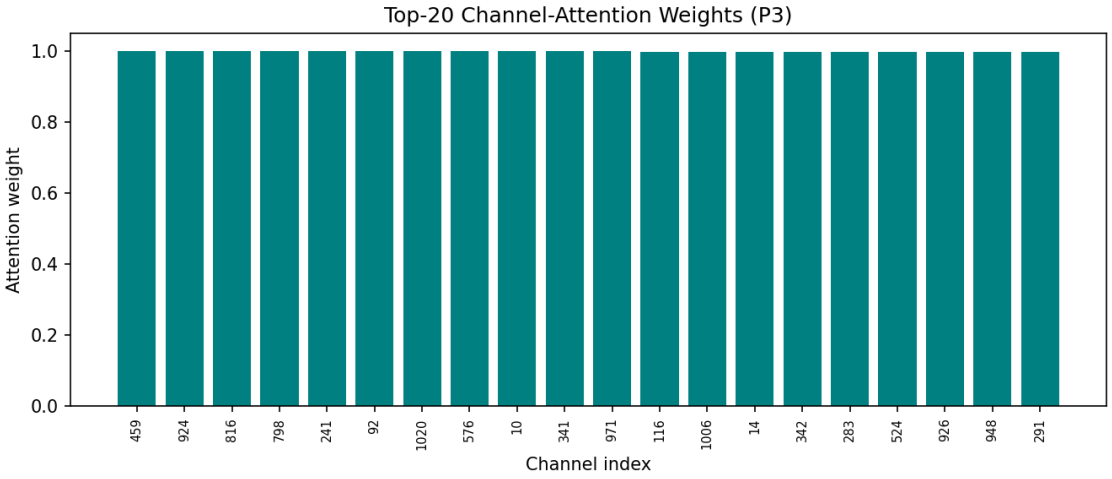


Figure 10: Top-20 channel-attention weights of the P_3 attention module.

Inference.

- All of the 20 largest weights are about 1.0, so the sigmoid is saturated for these channels and no small group of channels stands out.
- The plot shows only the top 20 of 1024 channels for one image, so we cannot tell how the other channels are weighted or whether the weights change from image to image.

28.2 Spatial-Attention Visualization

For spatial attention:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

Resize the attention map to:

$$224 \times 224$$

and overlay it on the input image.
Display:

Input | Spatial Attention Map | Attention Overlay

Students should identify the image regions receiving high attention values.

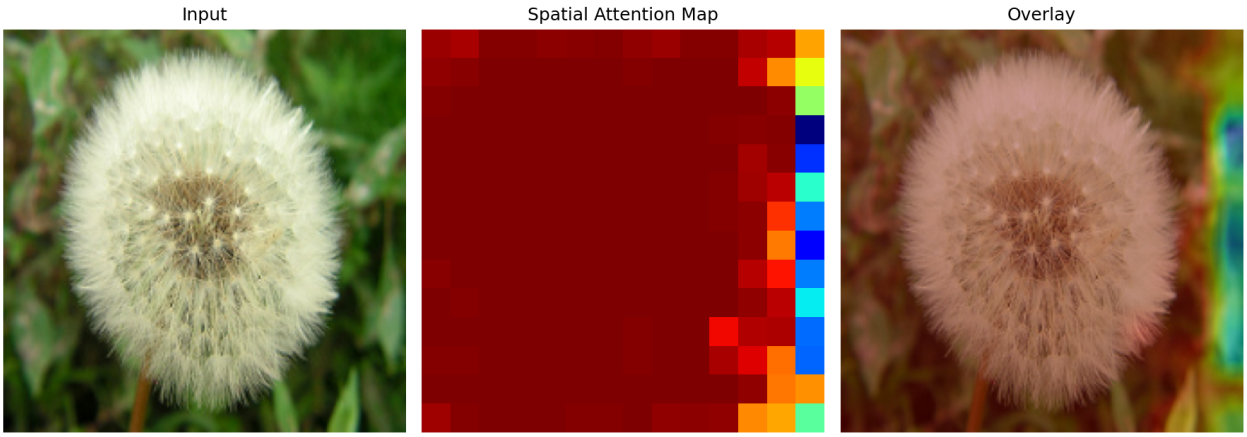


Figure 11: Input, spatial attention map (14×14 at P_3) and overlay for a dandelion image.

Inference.

- The map has almost the same high value over the whole 14×14 grid, on the flower and on the background, so it does not separate the dandelion from the leaves.
- The only change is in the right-most two or three columns, where the values drop. This looks like an edge effect of the convolution with padding and does not match any flower region.

28.3 Self-Attention Visualization

For self-attention, students may visualize one of the following:

- attention associated with a selected query location;
- mean attention across query positions;
- maximum attention across query positions; or
- an aggregated spatial self-attention response.

The visualization should be reshaped into a spatial form and, when appropriate, resized to the image resolution.

Students should analyze how distant regions interact within the feature map.

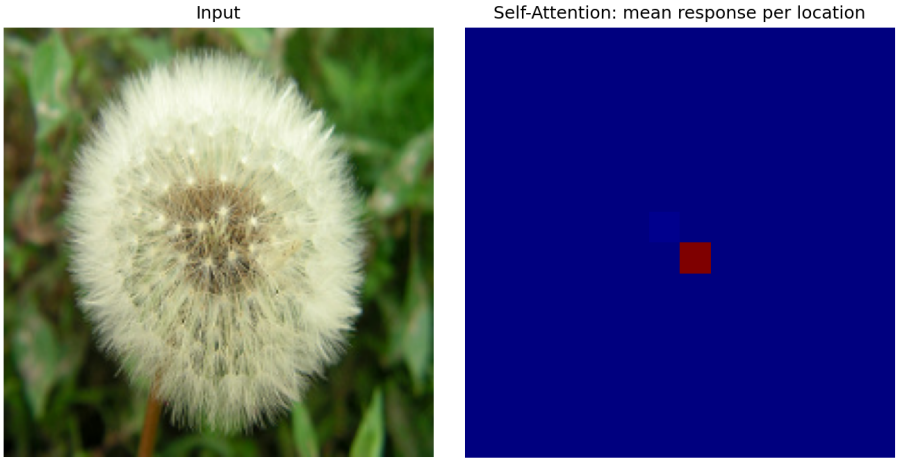


Figure 12: Input and mean self-attention response per location at P_3.

Inference.

- The mean response is concentrated on one location just below and to the right of the flower centre, and almost all other locations are near zero.
- For this image, self-attention does not show distant regions interacting; its attention is focused on a single position inside the flower. We would need more images to know if this is typical.

29. Attention Position Visualization

Use the same test image to visualize spatial attention at different ResNet50 stages.

Generate:

$$M_s^{P_2}, \qquad M_s^{P_3}, \qquad M_s^{P_4}.$$

Display the maps as:

$$\boxed{\text{Input} \mid M_s^{P_2} \mid M_s^{P_3} \mid M_s^{P_4}}$$

The objective is to visually study how attention changes as feature representations become deeper.

Possible observations may include:

- P_2 responding strongly to edges, colors, textures, and fine structures;
- P_3 responding to flower parts and intermediate object structures;
- P_4 responding to larger semantic flower regions.

These observations must be based on the student's experimental visualizations.

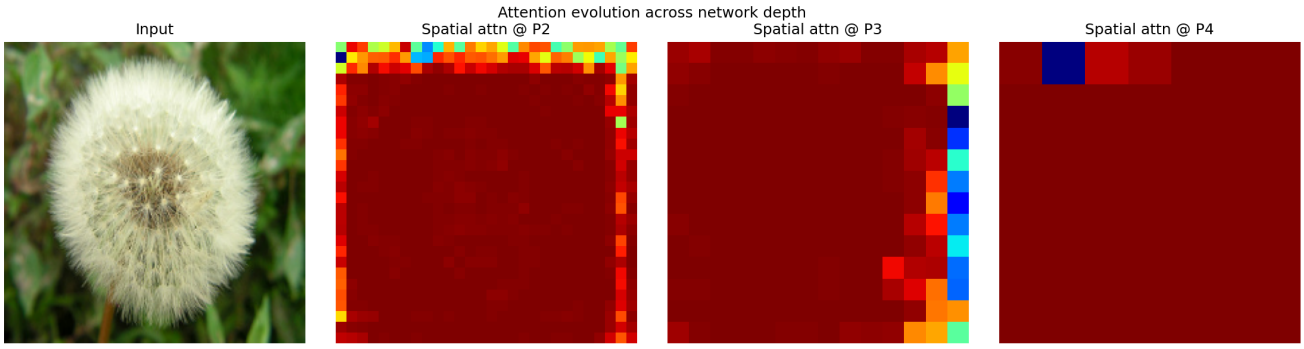


Figure 13: Spatial attention maps at P_2 (28×28), P_3 (14×14) and P_4 (7×7) for the same dandelion image. P_2 and P_4 are from Channel–Spatial models; P_3 is from the spatial-only model.

Inference.

- At P_2 the map is high and almost uniform inside the image and changes only at the border (top rows and left and right columns). At P_3 the only change is in the two right-most columns. At P_4 the map is uniform except for one low cell at the top.
- None of the three maps separates the flower from the background, and the variation that does appear is at the image borders. This is probably a zero-padding effect of the 7×7 convolution and not image content. The deeper maps look smoother only because the grids are smaller (28, 14 and 7 cells per side).
- So these maps do not show P_2 responding to edges and textures or P_4 responding to large flower regions. This is one image, and the P_3 map comes from the spatial-only model, so it is only a rough comparison.

30. Qualitative Comparison Across Attention Mechanisms

Select at least:

3 correctly classified images

and

2 incorrectly classified images.

For every selected image, compare the behavior of multiple models.
A recommended arrangement is:

Input Image
↓
Baseline Grad-CAM
Channel-Attention Grad-CAM
Spatial-Attention Map
Channel+Spatial Grad-CAM
Self-Attention Response

Students may alternatively arrange the images horizontally in the report.

30.1 Qualitative Criteria

The student should discuss:

- foreground localization;
- background suppression;
- localization of petals;
- localization of the flower center;
- activation compactness;
- activation coverage;
- incorrectly attended regions;
- differences between correctly and incorrectly classified samples;

- differences between shallow and deep attention; and
- consistency of attention across different images.

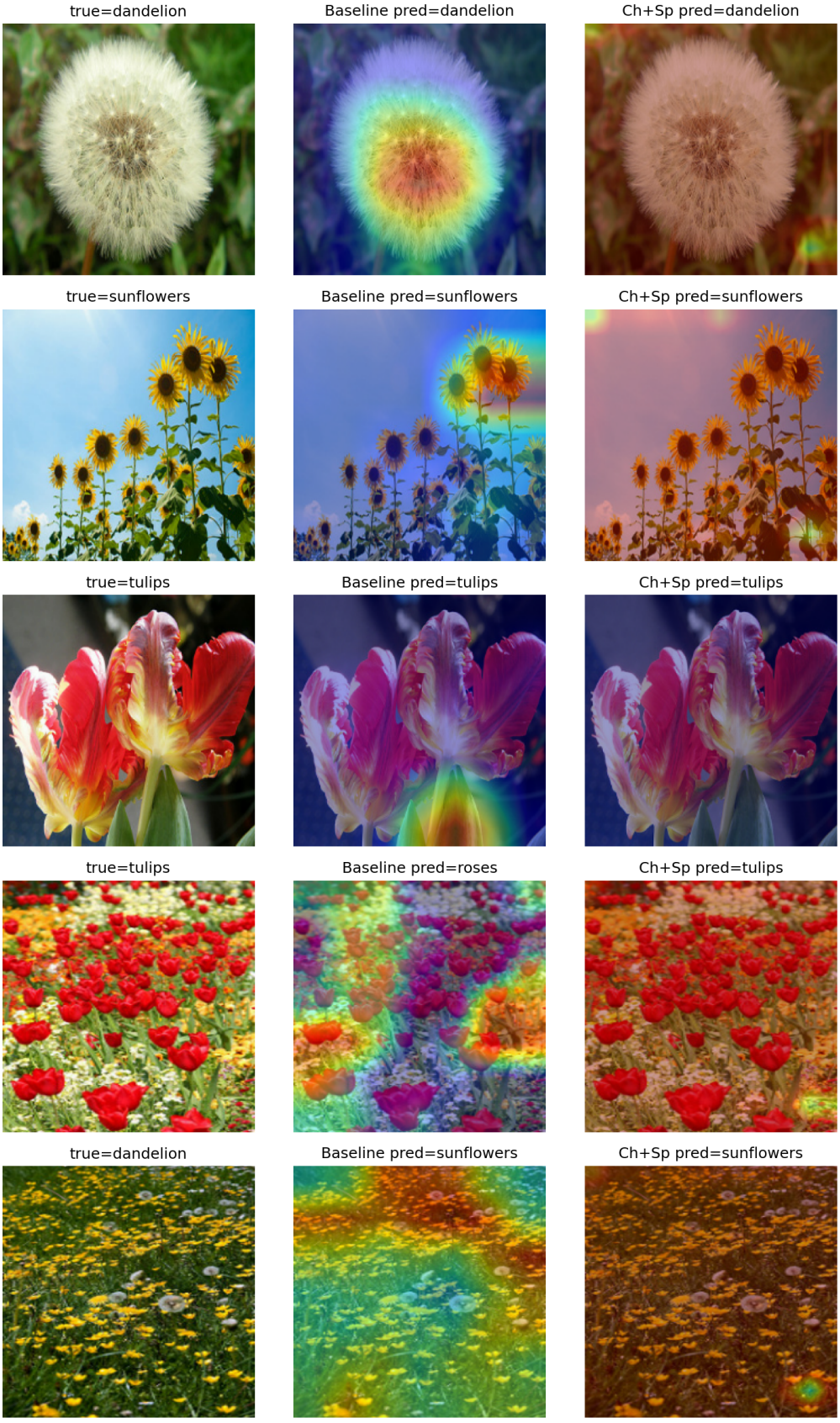


Figure 14: Qualitative comparison on five test images (three correct, two incorrect for the baseline): input, baseline Grad-CAM and Channel+Spatial (P_3) Grad-CAM with predicted labels.

Inference.

- **Correct images (baseline):** the baseline Grad-CAM is on the seed head for the dandelion, on the upper-right sunflowers and some nearby sky for the sunflower image, and on the stem and leaf at the bottom, not the petals, for the tulips. The baseline does not always use the flower itself as evidence.
- **Incorrect images (baseline):** the tulip field predicted as roses and the dandelion meadow predicted as sunflowers both give spread-out baseline activation over a cluttered scene with many small flowers, instead of one compact region.
- **Channel+Spatial:** its Grad-CAM is almost uniform, with a small spot at the bottom-right in four of the five images (and one at the top-left of the sunflower image). Because the spot is in the same place each time, it looks like a border effect. This map was also taken at the spatial-attention output layer, so it cannot be compared with the baseline maps and says nothing about localisation or background.
- **Predictions:** the attention model is right on the three images the baseline gets right and also fixes the tulip field (the baseline said roses), but both predict sunflowers for the dandelion meadow. On the whole test set each model is right on 14 images that the other misses (McNemar $p = 1.0$), so there is no overall difference.
- These five images agree with Section 29: the spatial attention maps show little image-dependent structure. Only five images were checked.

31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;
2. training and validation accuracy curves;
3. training and validation loss curves;
4. confusion matrix;
5. feature map before attention;
6. channel-attention weights;
7. spatial-attention heatmap;
8. spatial-attention overlay;
9. self-attention response for the self-attention experiment;
10. feature map after attention;
11. baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. attention maps at P_2, P_3, and P_4; and
14. at least one graphical comparison of the performances of different attention mechanisms.

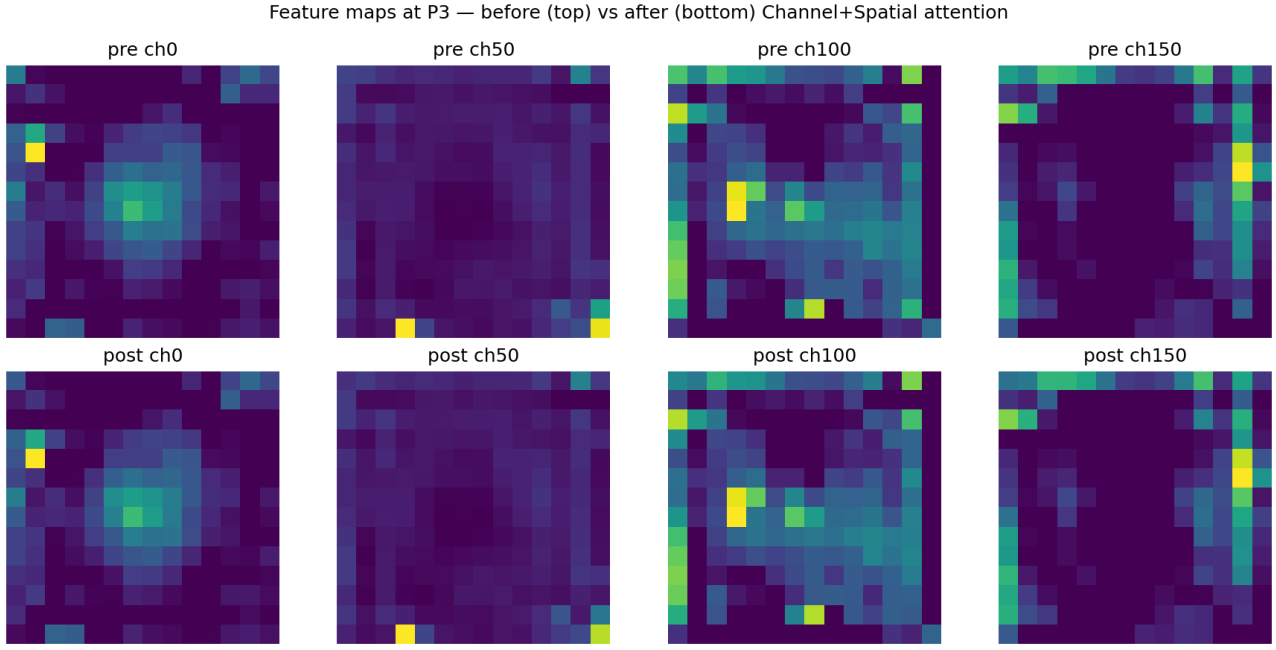


Figure 15: Feature maps at P_3 before (top) and after (bottom) Channel+Spatial attention, channels 0, 50, 100 and 150.

Inference.

- After attention the maps look almost the same as before. There are only small changes, for example at the top

right of channel 0 and in the right-hand column of channel 100.
- This fits with the attention weights being close to 1 and the spatial map being almost uniform: the module changes the features only a little, which matches Channel+Spatial (P_3) having the same test accuracy as the baseline (0.9328).

32. Recommended Visualization Layout

For one representative test image, students are encouraged to generate a figure in the following format:

Input	Baseline Grad-CAM	Channel Grad-CAM	Spatial Attention Map	Channel+Spatial Grad-CAM

For students completing the additional attention comparison, the figure may be extended to:

Input|Baseline|SE|ECA|CBAM|Self-Attention

using appropriate visualization methods.

33. Required Experimental Analysis

Quantitative result.

Self-attention at P_3 is the best model (accuracy 0.9437, precision 0.9428, recall 0.9446, macro F1 0.9435), followed by channel attention (0.9365 / 0.9359). The baseline scores 0.9328 / 0.9323, and spatial, ECA and CBAM are below it. For Channel–Spatial, P_3 is the best position (0.9328) compared with P_1 (0.9093), P_2 (0.9147) and P_4 (0.9183). All of these differences are within about ± 2 points on a 551-image test set.

Complexity result.

The extra parameters range from 5 (ECA) and 99 (spatial), through about 132 k (channel, CBAM, Channel–Spatial), up to 1.31 M (self-attention), on top of a 23.6 M baseline. Inference time is between 90 and 115 ms. Only self-attention (about 21% slower) is clearly slower; the other differences are within timing noise.

The final report should contain three forms of analysis.

33.1 Quantitative Analysis

Report:

Accuracy + Precision + Recall + Macro F1

33.2 Complexity Analysis

Report:

Parameters + Inference Time

33.3 Qualitative Analysis

Report:

Grad-CAM + Attention Maps + Visual Interpretation

Therefore, the overall evaluation should be:

Quantitative + Computational + Qualitative

Qualitative result.

On five test images the baseline Grad-CAM is on the flower head for the dandelion, on some of the flowers and sky for the sunflowers, on the stem for the tulips, and spread out for the two misclassified images. The Channel+Spatial Grad-CAM (taken at the spatial-attention output layer, so not comparable) is almost uniform with a spot in the bottom-right corner. The spatial attention maps at P_2, P_3 and P_4 are almost uniform and only change at the

borders, the channel weights are close to 1, the feature maps change little after attention, and self-attention focuses on one central location. This matches Channel+Spatial having the same accuracy as the baseline, but it is based on only a few images.

34. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:

- identical training images;
- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;
- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.

For example, when studying attention position:

$$\boxed{\text{Only Attention Position Should Change}}$$

and when studying attention type:

$$\boxed{\text{Only Attention Type Should Change.}}$$

35. Minimum Mandatory Ablation Set

If laboratory time is limited, every student must complete at least the following experiments:

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel+Spatial attention at P_3.
5. Channel+Spatial attention at P_2.
6. Channel+Spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of the classification performance of the attention models.

This minimum set allows students to study:

$$\boxed{\text{Attention Type}}$$

and

$$\boxed{\text{Attention Position}}.$$

36. Additional Student Exploration

Students interested in extended experimentation should select at least two mechanisms from:

$$\boxed{\text{SE,} \quad \text{ECA,} \quad \text{CBAM,} \quad \text{Coordinate Attention,} \quad \text{Self-Attention}}$$

and integrate them with the same ResNet50 backbone.

Students should maintain the same:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- epochs;
- fine-tuning strategy; and
- attention position.

The additional investigation should include both:

Quantitative Comparison

and

Qualitative Comparison.

37. Questions Students Must Answer

1. What is the purpose of residual connections in ResNet50? ans: residual connection will skip a block and is directly added into the output. this stops the gradients from vanishing in the deep learning networks
2. What does conv2_x represent compared with conv5_x? ans: conv2_x is closer to the input and operates at a higher spatial resolution (56×56), so it captures low-level features such as edges and colors. In contrast, conv5_x is closer to the output and operates at a lower spatial resolution (7×7), capturing richer, high-level semantic features
3. Why does the spatial resolution decrease with network depth? ans: due to convolutions and pooling in each layer, ther resolution decreases so that the netwoek build bigger receptive field which would learn large scale patterns
4. Why does the number of feature channels increase with network depth? ans: to compensate for the decreasing spatial detail, we add more channels so it can learn more complex and varied patterns
5. What is meant by low-level, intermediate, and high-level features? ans:Low-level features learn edges, colors, textures; intermediate level features learn parts and shapes; high-level features learn whole-object, class-related concepts
6. What is the purpose of an attention mechanism? ans: to let the network focus on the important features of a feature map instead of treating everingt equally
7. What does channel attention attempt to learn? ans: which feature channels matter most for the task
8. What does spatial attention attempt to learn? ans: which locations in the image matter most
9. How is channel attention different from spatial attention? Channel attention learns the most important features, spatial attention learns most important regions in a feature map, one works along the channel axis, the other along the height/width axes.
10. Why can channel and spatial attention be used sequentially? ans: channel attention tells which channel is important whole spatial attention tells where the important feature of the feature map lies
11. Why might channel attention behave differently at conv3_x and conv5_x? ans: Channels in `conv3_x` represent simpler, more local patterns, whereas channels in `conv5_x` represent more abstract concepts.
12. Why can spatial attention be useful at an intermediate ResNet stage? ans: even at the intermediate stage, the netwrok will have still a decent resolution to loacate important parts of a feature map
13. Why is self-attention computationally expensive when the feature map has high spatial resolution? ans: since self attentions campoares a location with every other location, its computatin increases with the matrix size, hence at high resolution, the neural netwrok becomex expensive and slow
14. What happens to the size of the self-attention matrix when the spatial resolution increases? ans: it grows exponentially at the rate of x power 4, since it will have 4 time more pixels, attention matrix become 16 times larger

15. How does attention performance vary when the insertion position is changed? ans: depends on both the dataset and the task, and even for same positions at each convolution layer will have different accuracy performance
16. How does attention at P_2 differ visually from attention at P_4? ans: at p2, the images looks shapr and focuses on details like edges and textures, at p4, the images becomes blurry but has a distinct boundary of what the flower shape is
17. Does applying attention at multiple stages always improve classification performance? ans: not necessary, more attention maps doesnt always gaurantee more accuracy
18. Does Channel→Spatial attention behave differently from Spatial→Channel attention? Yes, potentially. Channel-first picks important feature types before locating them spatially; spatial-first picks important regions before weighting feature types. The order can lead to slightly different refined features and different final accuracy — compare empirically.
19. How does the spatial-attention kernel size influence the attention map? ans: A larger kernel, such as 7×7 instead of 3×3, considers a wider neighborhood and can capture broader spatial context. A smaller kernel focuses on more local details but may miss wider spatial relationships.
20. How does SE attention differ from ECA attention? ans: Both mechanisms learn channel-wise weights. SE uses two fully connected layers to compute them, while ECA uses a lightweight one-dimensional convolution across channels. ECA therefore typically adds fewer parameters and computation.
21. How does CBAM combine channel and spatial attention? ans: CBAM applies channel attention first, then applies spatial attention to the channel-refined features. This is a sequential channel-to-spatial combination.
22. How does self-attention differ from channel attention? ans: Channel attention reweights each channel across all spatial locations. Self-attention allows individual spatial locations to exchange information directly, helping the model capture relationships between distant parts of the image.
23. Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy? ans: No. A larger parameter count does not guarantee higher accuracy. A lighter mechanism may achieve comparable or better performance, so accuracy should be considered alongside model complexity.
24. What is the difference between an internal attention map and Grad-CAM? ans: An internal attention map is produced by an attention module during the model's forward pass and is part of its computation. Grad-CAM is generated after a prediction by using gradients to highlight image regions that influence a selected class score; it does not alter the model's forward computation.
25. Why is Grad-CAM called a post-hoc explanation technique? ans: Grad-CAM is computed after the model makes a prediction to explain that prediction. It does not participate in the model's forward pass, unlike an attention module built into the network.
26. Does the attention model suppress irrelevant background regions? ans: It may, but this is not guaranteed. Inspect attention visualizations or Grad-CAM overlays across several test images to determine whether the model focuses more on the flower and less on irrelevant background regions.
27. Are the attention maps of correctly classified and incorrectly classified images different? ans: They can differ. Correct predictions may show focus on discriminative parts of the flower, while incorrect predictions may show diffuse attention or focus on misleading regions. Compare multiple examples, since this pattern is not guaranteed for every image.
28. Which flower classes benefit most from attention? ans: Determine this from the class-wise metrics and confusion matrices. Classes that the baseline frequently confuses may benefit if attention helps the model focus on features that distinguish them.
29. What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone? ans: Accuracy provides a single overall score, whereas a confusion matrix shows which classes are being mistaken for one another. It can reveal class-specific strengths and weaknesses that overall accuracy hides.
30. Why is macro F1 useful in addition to accuracy? ans: Accuracy can obscure poor performance on less frequent classes. Macro F1 computes the F1 score for each class and averages them equally, giving a clearer view of class-wise performance, especially when the classes are imbalanced.

31. What is meant by the accuracy–complexity trade-off? ans: It is the balance between predictive accuracy and the resources a model requires, such as parameters and computation. A slightly less accurate model may be preferable if it is substantially smaller or faster.
32. What is meant by the F1–inference-time trade-off? ans: It is the balance between macro F1 and the time required to make predictions. A model with a slightly lower F1 score may be preferable when it provides substantially faster inference.
33. Why must every ablation study use the same dataset split? ans: Using the same samples in each split helps ensure that performance differences are due to the model change being tested, rather than differences in the training, validation, or test data.
34. Why must the same training protocol be maintained when comparing different attention mechanisms? ans: Keeping the training protocol fixed isolates the effect of the attention mechanism. If other factors, such as the number of epochs or learning rate, also change, it becomes difficult to determine what caused a performance difference.
35. What conclusions can be drawn from the Grad-CAM comparison? ans: Compare whether the models focus on the flower, suppress irrelevant background regions, or attend to different areas. Interpret these visualizations together with quantitative results and across several representative test images; Grad-CAM alone does not prove that an attention module caused a performance change.

38. Observations to be Recorded

Students should record the following during experimentation:

1. Input and output feature dimensions for each attention module.
2. Training accuracy.
3. Validation accuracy.
4. Testing accuracy.
5. Training loss.
6. Validation loss.
7. Macro precision.
8. Macro recall.
9. Macro F1-score.
10. Number of trainable parameters.
11. Total number of model parameters.
12. Average inference time.
13. Confusion matrix.
14. Channel-attention weights.
15. Spatial-attention maps.
16. Self-attention responses.
17. Grad-CAM maps.
18. Attention maps for different insertion positions.

Recorded observations.

Experimental configuration. ResNet50 (ImageNet) backbone with the Adam optimizer. The learning rate was 10^{-3} for 3 epochs with the backbone frozen, then 10^{-4} for 2 epochs with the last stage (conv5_x) unfrozen. Loss: categorical cross-entropy, batch size 16, dropout 0.3, random seed 42, early stopping on validation accuracy (patience 5, best weights restored). The split was the same 70:15:15 split for every run (2,569 / 550 / 551 images) and training was done on a Colab GPU. Inference time is the average of 50 single-image forward passes. The notebook was run in its `QUICK_RUN` mode (5 epochs in total) and not the full 25-epoch setting, so all results come from a short training schedule.

Item 1 – Input and output feature dimensions of each attention module. All attention modules preserve the tensor shape.

Position	Input $H \times W \times C$	Output $H \times W \times C$
P_1 (after conv2_x)	$56 \times 56 \times 256$	$56 \times 56 \times 256$
P_2 (after conv3_x)	$28 \times 28 \times 512$	$28 \times 28 \times 512$
P_3 (after conv4_x)	$14 \times 14 \times 1024$	$14 \times 14 \times 1024$
P_4 (after conv5_x)	$7 \times 7 \times 2048$	$7 \times 7 \times 2048$

Items 2, 3, 5, 6 – Final-epoch (epoch 5) training and validation accuracy and loss. "–" means the training history of that run was not saved (B1 and CBAM).

Model	Train acc.	Val acc.	Train loss	Val loss
A0 Baseline	0.9677	0.9364	0.0970	0.2210
A1 Channel (P_3)	0.9611	0.9109	0.1190	0.2784
A2 Spatial 7×7 (P_3)	0.9611	0.9200	0.1075	0.2406
A3 Channel–Spatial (P_3)	0.9642	0.9182	0.1174	0.2349
A4 Self-Attention (P_3)	0.9681	0.9218	0.0966	0.2584
A5 Combined (P_3)	0.9560	0.9218	0.1318	0.2381
B1 Channel–Spatial (P_1)	–	–	–	–
B2 Channel–Spatial (P_2)	0.9591	0.9018	0.1167	0.3094
B4 Channel–Spatial (P_4)	0.9728	0.9182	0.0922	0.3130
C Channel–Spatial (P_2+P_3+P_4)	0.9587	0.9073	0.1277	0.2997
C Channel–Spatial (P_3+P_4)	0.9673	0.9145	0.1012	0.2981
D Spatial→Channel (P_3)	0.9580	0.9345	0.1203	0.2269
ECA (P_3)	0.9603	0.9164	0.1054	0.2721
CBAM (P_3)	–	–	–	–
Spatial 3×3 (P_3)	0.9607	0.8909	0.1241	0.3917
Spatial 5×5 (P_3)	0.9720	0.9182	0.0895	0.2292

Items 4, 7–12 – Test accuracy, macro precision, macro recall, macro F1, parameters and inference time. Inference time is average ms per image over 50 single-image passes; "–" = not recorded.

Model	Test acc.	Precision	Recall	Macro F1	Trainable params	Total params	Time (ms)
A0 Baseline	0.9328	0.9358	0.9305	0.9323	14,986,245	23,597,957	95.46
A1 Channel (P_3)	0.9365	0.9363	0.9377	0.9359	15,118,405	23,730,117	90.41
A2 Spatial 7×7 (P_3)	0.9256	0.9304	0.9223	0.9248	14,986,344	23,598,056	109.36
A3 Channel–Spatial (P_3)	0.9328	0.9341	0.9312	0.9316	15,118,504	23,730,216	96.77
A4 Self-Attention (P_3)	0.9437	0.9428	0.9446	0.9435	16,298,246	24,909,958	115.30
A5 Combined (P_3)	0.9347	0.9375	0.9319	0.9340	16,430,505	25,042,217	97.22
B1 Channel–Spatial (P_1)	0.9093	0.9102	0.9095	0.9085	14,994,808	23,606,520	106.76
B2 Channel–Spatial (P_2)	0.9147	0.9187	0.9144	0.9140	15,019,656	23,631,368	104.28
B4 Channel–Spatial (P_4)	0.9183	0.9202	0.9147	0.9166	15,512,808	24,124,520	105.93
C Channel–Spatial (P_2+P_3+P_4)	0.9256	0.9266	0.9270	0.9249	15,678,478	24,290,190	94.10
C Channel–Spatial (P_3+P_4)	0.9020	0.9062	0.9032	0.9013	15,645,067	24,256,779	97.22
D Spatial→Channel (P_3)	0.9310	0.9333	0.9282	0.9299	15,118,504	23,730,216	114.68
ECA (P_3)	0.9111	0.9105	0.9117	0.9106	14,986,250	23,597,962	103.11
CBAM (P_3)	0.9074	0.9064	0.9060	0.9059	15,118,504	23,730,216	106.64
Spatial 3×3 (P_3)	0.9165	0.9210	0.9131	0.9157	14,986,264	23,597,976	91.20
Spatial 5×5 (P_3)	0.9129	0.9135	0.9114	0.9119	14,986,296	23,598,008	103.93

Item 13 – Confusion matrix. Recorded for the baseline and Channel–Spatial (P_3) on the 551 test images (126 dandelion, 98 daisy, 118 tulips, 111 sunflowers, 98 roses); both classify 514 correctly (see Section 25). Baseline rows/columns (dandelion, daisy, tulips, sunflowers, roses): [120,1,2,3,0], [0,89,5,3,1], [0,0,114,0,4], [2,2,4,103,0], [1,1,7,1,88]. Channel–Spatial (P_3): [116,6,0,4,0], [1,92,3,2,0], [0,0,116,0,2], [3,2,0,105,1], [1,2,9,1,85].

Item 17 – Grad-CAM maps. Done for five test images (three correct and two incorrect for the baseline), for the baseline and Channel–Spatial P_3. See Sections 27.1 and 30.

Items 14, 15, 16 – Channel-attention weights, spatial-attention maps and self-attention responses. Done for one dandelion test image at P_3 (Sections 28.1, 28.2 and 28.3). The top-20 channel weights are all about

1.0, the spatial map is almost uniform except at the right border, and the self-attention response is concentrated on one central location.

Item 18 – Attention maps for different insertion positions. Done for one dandelion image at P_2, P_3 and P_4 (Section 29). The maps are almost uniform and only change at the image borders.

39. Suggested Laboratory Report Structure

The laboratory report may be organized as follows:

1. **Aim**
2. **Dataset Description**
3. **ResNet50 Architecture**
4. **Attention Mechanisms**
5. **Proposed Experimental Architecture**
6. **Experimental Setup**
7. **Attention-Type Ablation**
8. **Attention-Position Ablation**
9. **Performance Results**
10. **Complexity Analysis**
11. **Training and Validation Graphs**
12. **Confusion Matrix Analysis**
13. **Grad-CAM Analysis**
14. **Internal Attention-Map Analysis**
15. **Qualitative Comparison**
16. **Additional Attention-Mechanism Comparison**
17. **Discussion**
18. **Conclusion**

40. Discussion Guidelines

Students should not merely reproduce numerical results.

The discussion should explain:

- why a particular attention mechanism produced a change in performance;
- how attention location influences feature refinement;
- how feature resolution changes attention behavior;
- how model complexity changes with different mechanisms;
- whether performance improvements justify additional computational cost;
- whether attention maps correspond to meaningful object regions;
- whether attention helps suppress the background;
- whether attention behaves differently for incorrect predictions; and
- whether quantitative and qualitative observations support each other.

Students should avoid making conclusions based on a single image.

Qualitative observations should preferably be based on several representative test images.

Discussion.

Why performance changed. Self-attention gave the only clear gain (+1.09 points). It lets all 196 positions of the 14×14 map interact, although for the example image its response collapsed onto one location. The cheaper modules changed accuracy by less than the error of the test set, and ECA and CBAM ended below the baseline. A newly initialised module in the middle of a pretrained network, trained for only five epochs, can slightly disturb the pretrained features.

Position and resolution. For Channel–Spatial attention, P_3 was the best position and P_1 the worst. At P_1 and P_2 the features are low-level, and at P_4 the map is only 7×7, which may explain part of this, but we did not test it. Using attention at several stages (P_3+P_4, P_2+P_3+P_4) did not improve over P_3 alone.

Cost against benefit. Channel attention costs 0.56% more parameters for +0.36 points. Self-attention costs 5.6% more parameters and about 21% more inference time for +1.09 points. Bigger models were not systematically better.

Background and wrong predictions. None of the spatial attention maps (P_2, P_3, P_4) separates the flower from the background, and the only structure in them is at the image borders. The attention model's Grad-CAM was taken at a different layer from the baseline's, so we cannot judge background suppression. For the baseline, the two misclassified images (fields of many small flowers) give spread-out activation, while the correct close-ups give one compact region. With five images this is an observation and not a conclusion.

Reliability. Every model was trained once and tested on 551 images (one image is 0.18 points), so the rankings are only rough. Repeated runs with different seeds would be needed to confirm them.

41. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.
Its effectiveness depends on:

$$\boxed{\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity.}\end{aligned}}$$

Students should also understand that classification performance alone does not provide a complete explanation of an attention mechanism.
A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

42. References

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LAB 8- github link